

Harihara Yashwanth Veldanda

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TECHNICAL SKILLS

Languages: Python, SQL, R, C, C++, HTML, CSS, Javascript
Frameworks and Libraries: Pandas, Numpy, ScikitLearn, Tensorflow, Matplotlib, Pytorch, Keras, NodeJS, React
Databases & Cloud: Google BigQuery, Azure, AWS, MySQL, PostgreSQL
Tools: Microsoft Office, VSCode, Jupyter, PowerBI, Tableau, ApacheSpark

PROFESSIONAL EXPERIENCE

- MathCo** **Bengaluru, India**
Data Analyst *October 2023 - July 2024*
- Partnered with Customers, Insights and Strategies team for **WALMART** (client) and executed incremental analysis, delivered insights that drove a 10% increase in overall customer spending and acquisition of 30% new Walmart+ members.
 - Diagnosed possible causes of Walmart+ member churn by analyzing customer sentiment, leveraged NLP to summarize customer pain points and optimized customer engagement through data analysis, reducing member cancellations by 20%.
 - Applied KNN for customer audience segmentation, enabling targeted strategies that led to a 30% increase in customer engagement.
 - Amplified Walmart+ member retention by 40% through predictive analytics on optimal engagement timings and behavioral patterns.
 - Piloted and evaluated 15+ A/B tests to gauge the impact of offers and promotions within retail industry, providing actionable insights to enhance marketing strategies leading to a 15% increase in member conversion rates.
 - Wrote complex SQL queries in BigQuery and performed data validation to ensure data reliability across datasets and analyses.
 - Developed insights, dashboards (Power BI/Tableau), and data-driven narratives, partnering with stakeholders to translate findings into actionable recommendations, contributing to up to 30% improvement in customer engagement and 15% increase in conversion rates.
 - Tech Stack: Python, SQL, Google BigQuery, PySpark
- MathCo** **Bengaluru, India**
Intern Analyst *January 2023 - June 2023*
- Designed predictive models for luxury fashion, identifying 90% of key sales drivers, boosting sales by 10%.
 - Constructed a simulator for stock estimation with a MAPE of 12%, using PowerBI and internal tools.
 - Streamlined data pipelines with Azure Data Factory, integrating machine learning models for enhanced insights.

PERSONAL PROJECTS

- Applied AI Engineering – Tyche Labs| LekhaAI** *May 2025 - Present*
- Fine-tuned a FLUX diffusion model using LoRA on ~1,500 image-prompt pairs to improve layout quality and text rendering in generated invitations.
 - Ran iterative experiments on prompt structure, resolution bucketing, and guidance scaling, improving visual fidelity by ~25% and reducing text artifacts by ~30% compared to the base model.
 - Deployed containerized GPU inference via Azure ML Online Endpoints (A100), optimizing cold-start latency and model loading.
 - Integrated GPT-4.0 nano to structure user prompts and handle conversational refinement before image generation.
 - Developed the full-stack pipeline (React + TypeScript frontend, Azure Functions backend) to support real-time generation and template preview.
 - Deployed the application to production with CI/CD(GitHub + Vercel) and custom domain configuration.
 - Demo: https://huggingface.co/tyche-labs/lekhaAI_flux_v1, www.lekhaai.com – Signup => Try AI now
 - Tech Stack: Python, PyTorch, Diffusers, Azure ML, Docker, REST APIs, React, TypeScript

ACADEMIC PROJECTS

- Retrieval-Augmented Generation (RAG) Assistant for Workday HR Workflows** *January 2025 - May 2025*
- Built a containerized RAG assistant indexing 10K+ HR records via RocksDB and FAISS for secure, low-latency retrieval.
 - Boosted query accuracy by 83% using context-grounded semantic search (cosine similarity < 0.5).
 - Automated common HR queries and personalized communications, reducing manual support workload by an estimated 40%.
 - Tech Stack: Python, SQL, RocksDB, Docker, FastAPI
- Construction of Medical Chatbot using Knowledge Graph in Neo4j** *August 2022 - December 2022*
- Designed a Neo4j knowledge graph linking diseases, symptoms, and treatments for intelligent query handling.
 - Integrated NLP-based entity extraction and intent detection, improving query accuracy by 25%.
 - Tech Stack: Python, Neo4j, Streamlit, Docker, RestAPI, NLTK, Scikit-learn
- Sentiment Analysis on Twitter data** *June 2022 - July 2022*
- Formulated a Sentiment Analysis project on 500,000+ Twitter posts using Python. Employed sophisticated techniques including lemmatization, stemming, and tf-idf calculations for data preprocessing.
 - Deployed SVC for dimensional reduction and Logistic Regression for sentiment analysis, achieving an f1 score of 0.59 for accurate and robust classification.
 - Tech Stack: Python, scikit-learn, NLTK, TF-IDF

EDUCATION

- Arizona State University** *Aug 2024 - May 2026*
- Master of Science - Data Science and Analytics Engineering* *GPA : 3.87*
 - Relevant Coursework: Artificial Intelligence, Timeseries Forecasting, Statistical Machine Learning, Data Mining*